

Diabetes Prediction using Stratified K-Fold Cross Validation

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Project Overview

This project demonstrates the use of **Stratified K-Fold Cross Validation** for evaluating machine learning models on the Diabetes dataset.

Rather than relying on a single train-test split, the project compares multiple validation strategies and shows how K-Fold Cross Validation provides a more stable and reliable estimate of model performance.

A Streamlit web application was also developed to provide an interactive interface for diabetes prediction.

Live Application

Diabetes Prediction Web App

Objectives

- Predict diabetes using patient medical measurements.
- Demonstrate the limitations of a single train-test split.
- Apply Stratified K-Fold Cross Validation.
- Compare multiple machine learning algorithms.
- Study the effect of different values of K on model performance.

Dataset

The project uses the Pima Indians Diabetes Dataset.

Input Features

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Diabetes Pedigree Function

- Age

Target Variable

- Outcome (0 = Non-Diabetic, 1 = Diabetic)

Data Preprocessing

Several medical attributes contained zero values that represent missing measurements.

The following columns were cleaned:

- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI

Missing values were replaced using the median of the respective feature.

Feature scaling was performed using StandardScaler.

Machine Learning Models

Three classification algorithms were evaluated:

1. Logistic Regression
2. Decision Tree Classifier
3. K-Nearest Neighbors (KNN)

Methodology

1. Single Train-Test Split

The dataset was repeatedly split using different random seeds.

Results showed that accuracy changes significantly depending on the random split, making evaluation unreliable.

2. Stratified K-Fold Cross Validation

To obtain a more reliable estimate:

- Stratified K-Fold was used.
- Number of folds (K) = 5.
- Shuffle = True.
- Random State = 42.

Stratification ensures that each fold preserves the original class distribution of diabetic and non-diabetic patients.

3. Pipeline-Based Evaluation

A Scikit-Learn Pipeline was used:

1. StandardScaler
2. Classification Model

This prevents data leakage by performing scaling independently within each fold.

Model Comparison

The models were evaluated using:

- Mean Accuracy
- Standard Deviation
- Cross-Validation Scores across all folds

Performance comparison was visualized using bar charts with error bars representing variability across folds.

Effect of K on Performance

The project also investigates how different values of K affect model evaluation.

Values tested:

$$K = \{2, 3, 5, 10, 20\}$$

For each value of K:

- Mean Accuracy was computed.
- Standard Deviation was computed.
- Results were visualized using error-bar plots.

Key Findings

1. Single train-test split results vary significantly.
2. Stratified K-Fold provides a more stable estimate of model performance.
3. Preserving class distribution is important for imbalanced healthcare datasets.
4. Logistic Regression achieved competitive and consistent performance.
5. $K = 5$ and $K = 10$ provide a good balance between bias and variance.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-Learn
- Matplotlib
- Seaborn
- Streamlit

Project Workflow

1. Load Diabetes Dataset
2. Perform Data Cleaning
3. Handle Missing Values
4. Standardize Features
5. Train Multiple Models
6. Apply Stratified K-Fold Cross Validation
7. Compare Performance Metrics
8. Deploy Interactive Streamlit Application

Web Application

The deployed Streamlit interface allows users to enter medical information and obtain an instant diabetes prediction.

Open Diabetes Prediction App

Conclusion

This project demonstrates the importance of proper model evaluation in healthcare machine learning applications. By replacing a single train-test split with Stratified K-Fold Cross Validation, the model evaluation becomes more robust, reliable, and representative of real-world performance.